

Second Normal Form (2NF) -- Relational Database Design

Definition: A relation is in Second Normal Form (2NF) if it is in 1NF and no non-prime attribute is partially dependent on any candidate key of the relation. Every non-key attribute must be fully functionally dependent on the primary key.

1. Detailed Technical Explanation

Functional Dependency Concepts:

* Full Functional Dependency: $X \rightarrow Y$ is a full functional dependency if removal of any attribute A from X means the dependency no longer holds.

Conversion Example: 1NF to 2NF

Relation R(Student_ID, Course_ID, Student_Name, Course_Fee)

Decomposition into 2NF:

1. STUDENT (Student_ID, Student_Name)

2. Core Concepts & Memory Keywords

- * Non-prime Attribute: An attribute that is not part of any candidate key.

3. Must-Write Points for Exams

- * 2NF is only relevant when the relation has a composite candidate key (candidate key containing 2 or more attributes).

4. Quick Recall Flow

1NF Relation -> Identify Partial Dependencies (Non-Prime -> Part of Key) -> Decompose Tables -> 2NF Achieved